

1 Solution — GIAM 3.4

Problem 7

1.1 Problem

True or false: the product of any two irrational numbers is irrational. Prove your answer.

1.2 Answer

False. $\sqrt{2} \cdot \sqrt{2} = 2.$
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1.3 Disproof

Disproof. Take $x = y = \sqrt{2}$.

By Theorem 1.6 of the text, $\sqrt{2}$ is irrational, so both x and y are irrational numbers. Their product is

$$x \cdot y = \sqrt{2} \cdot \sqrt{2} = \left(\sqrt{2}\right)^2 = 2,$$

and $2 = 2/1$ is rational.

So there exist irrational numbers whose product is rational, and the claim is false.



Nothing in the problem requires the two numbers to be different, and taking them equal makes this the shortest disproof in the section: only one irrationality fact is needed, and no closure argument at all.

Other witnesses, if a pair of distinct numbers is preferred:

$$\sqrt{2} \cdot \sqrt{8} = 4, \quad \sqrt{2} \cdot \sqrt{18} = 6, \quad \sqrt[3]{2} \cdot \sqrt[3]{4} = 2.$$

1.4 The Picture

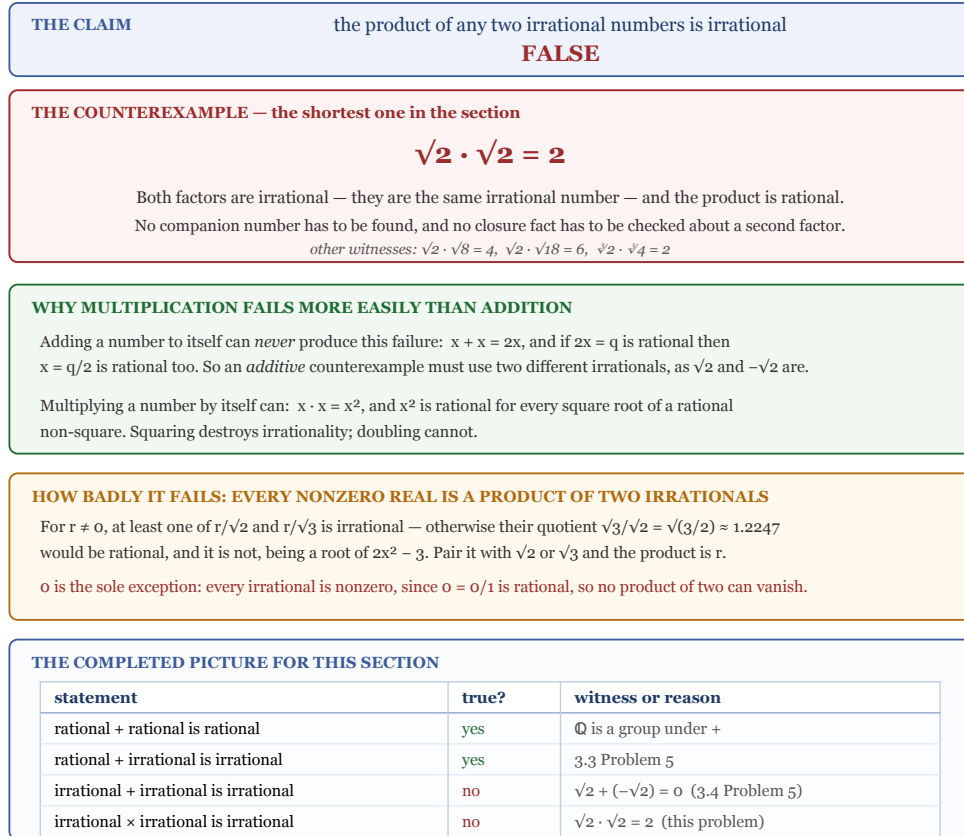


Figure 1.1: A diagram in five bands. The first states the claim — the product of any two irrational numbers is irrational — and the verdict, FALSE. The second gives the counterexample, the shortest in the section: root 2 times root 2 equals 2. Both factors are irrational, being the same irrational number, and the product is rational; no companion number has to be found and no closure fact has to be checked about a second factor. Other witnesses are root 2 times root 8 equals 4, root 2 times root 18 equals 6, and cube root of 2 times cube root of 4 equals 2. The third band explains why multiplication fails more easily than addition: adding a number to itself can never produce this failure, since x plus x equals $2x$, and if $2x$ equals a rational q then x equals q over 2 is rational too — so an additive counterexample must use two different irrationals, as root 2 and minus root 2 are. Multiplying a number by itself can, since x times x equals x squared, and x squared is rational for every

square root of a rational non-square. Squaring destroys irrationality; doubling cannot. The fourth band shows how badly it fails: every nonzero real is a product of two irrationals. For r nonzero, at least one of r over root 2 and r over root 3 is irrational, otherwise their quotient root 3 over root 2, which is the square root of three halves, about 1.2247, would be rational, and it is not, being a root of $2x$ squared minus 3. Pair it with root 2 or root 3 and the product is r . Zero is the sole exception, because every irrational is nonzero — 0 equals 0 over 1 is rational — so no product of two irrationals can vanish. The fifth band gives the completed picture for the section: rational plus rational is rational, true, because the rationals form a group under addition; rational plus irrational is irrational, true, by 3.3 Problem 5; irrational plus irrational is irrational, false, since root 2 plus minus root 2 equals 0, from 3.4 Problem 5; and irrational times irrational is irrational, false, since root 2 times root 2 equals 2, from this problem.

1.5 Remark — Why This Is the Easiest Counterexample of the Three

Compare the three disproofs in this run of exercises. Each needs a witness, but they cost different amounts:

claim	witness	what must be checked
irrational + irrational is irrational	$\sqrt{2}, -\sqrt{2}$	$\sqrt{2}$ irrational, $-\sqrt{2}$ irrational
irrational $\times$ irrational is irrational	$\sqrt{2}, \sqrt{2}$	$\sqrt{2}$ irrational

The additive case needs a second lemma — that the negative of an irrational is irrational — because the two summands *cannot* be the same number. The multiplicative case needs nothing beyond the irrationality of $\sqrt{2}$, because the two factors can be.

That asymmetry is not an accident of the choice of witness; it is forced.

1.6 Remark — Doubling Cannot Fail, Squaring Can

Here is why the additive counterexample is obliged to use two distinct numbers.

Observation. For a real x , the sum $x + x$ is rational if and only if x is rational.

Proof. If x is rational so is $2x$. Conversely if $2x = q \in \mathbb{Q}$, then $x = q/2$, and $\mathbb{Q}$ is closed under division by 2, so x is rational. ■

Doubling is an *invertible* operation on $\mathbb{Q}$ — you can undo it and stay inside — so it cannot carry an irrational into the rationals. Squaring is not invertible on $\mathbb{Q}$: the square root of a rational need not be rational, so $x^2 \in \mathbb{Q}$ does **not** force $x \in \mathbb{Q}$, with $x = \sqrt{2}$ the standard witness. Every rational non-square supplies another — $\sqrt{3}, \sqrt{5}, \sqrt{7}$ all square into $\mathbb{Q}$ while lying outside it.

So the failure here is precisely the failure of square roots to preserve rationality — the very fact that made $\sqrt{2}$ interesting in Chapter 1. The counterexample to this problem is the irrationality of $\sqrt{2}$, restated.

1.7 Remark — Products Need Not Be Rational Either

As with the additive version, disproving the claim does not establish its opposite. The product of two irrationals can be either:

$$\sqrt{2} \cdot \sqrt{2} = 2 \text{ (rational),} \quad \sqrt{2} \cdot \sqrt{3} = \sqrt{6} \approx 2.4495 \text{ (irrational).}$$

The irrationals are $\mathbb{R} \setminus \mathbb{Q}$, a complement rather than a structure, so they are closed under nothing. What *is* true is the mixed statement, which is 3.3 Problem 7 read forwards: a **non-zero** rational times an irrational is irrational. The “non-zero” is essential there and has no additive counterpart, since $0 \cdot \sqrt{2} = 0$ but $0 + \sqrt{2} = \sqrt{2}$.

That is the one place the multiplicative story is *more* delicate: $(\mathbb{Q}, +)$ is a group outright, whereas multiplicatively one must discard 0 first.

1.8 Remark — Every Non-Zero Real Is a Product of Two Irrationals

The failure is as total as it can be, with exactly one exception.

Claim. Every real $r \neq 0$ is a product of two irrational numbers, and 0 is not.

Proof. Let $r \neq 0$. At least one of $r/\sqrt{2}$ and $r/\sqrt{3}$ is irrational: if both were rational, so would be their quotient

$$\frac{r/\sqrt{2}}{r/\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{2}} = \sqrt{\frac{3}{2}} \approx 1.2247,$$

which is a root of $2x^2 - 3$ and has no rational value. If $r/\sqrt{2}$ is irrational, write $r = (r/\sqrt{2}) \cdot \sqrt{2}$; otherwise $r = (r/\sqrt{3}) \cdot \sqrt{3}$.

For 0 : every irrational number is non-zero, since $0 = 0/1$ is rational. A product of two non-zero reals is non-zero, so 0 is not such a product. ■

Compare Problem 5, where **every** real without exception is a sum of two irrationals. Multiplication has a single blind spot at 0 , for the same reason the “non-zero” caveat appears above.

1.9 Remark — The Section Completed

Taken together, Problems 5, 6 and 7 of this section, with 3.3 Problems 5 and 7, settle every combination:

statement	true?	witness or reason
rational + rational is rational	yes	$(\mathbb{Q}, +)$ is a group
rational $\times$ rational is rational	yes	$\mathbb{Q}$ closed under $\times$
rational + irrational is irrational	yes	3.3 Problem 5
non-zero rational $\times$ irrational is irrational	yes	3.3 Problem 7
irrational + irrational is irrational	no	$\sqrt{2} + (-\sqrt{2}) = 0$
irrational $\times$ irrational is irrational	no	$\sqrt{2} \cdot \sqrt{2} = 2$

The rule that emerges: a statement is a theorem exactly when at least one operand comes from the structured set $\mathbb{Q}$. Two irrationals are two unconstrained objects, and there is nothing to stop their irrational parts cancelling — additively via x and $-x$, multiplicatively via x and $1/x$, or in this case via x and x itself.